

Problem Set - 5

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1 Mechanics

1.1 Chase in Fog

A pirate ship is standing still in the middle of the ocean. Suddenly, another ship passes right next to it with velocity v . After a short time T , the pirate ship starts chasing the other ship with constant speed u . Because of the fog, the pirates can see the other ship when they are right next to it, i.e. when they are virtually on the same point. Consider the ships as point objects. What is the optimal trajectory the pirates need to follow to catch the other ship in minimum time. With this trajectory, what is the interval for the time $t_{min} \leq t \leq t_{max}$ it takes them to catch the ship?

1.2 Thread Around a Cylinder

One end of a thread is tied into a loop of length $L > 2\pi R$, and a cylinder of radius R is put through the loop. The coefficient of friction between the thread and the cylinder is μ . The free end of the thread is being pulled parallel to the axis of the cylinder (as shown by arrow in the photo) while keeping the cylinder at rest. If the length of the loop is longer than a critical value, $L > L_0$, the loop can slide along the cylinder without changing its shape, otherwise the friction “locks” it into a place and increasing the pulling force would eventually just break the thread. Find this critical value L_0 . The weight of the thread is to be neglected; the thread will not twist when being pulled. (EuPho 2021, T2)

Hint: It might be useful to know that

$$2 \int \sqrt{1+x^2} dx = x\sqrt{1+x^2} + \sinh^{-1} x$$

where $\sinh^{-1} x = \ln(x + \sqrt{1+x^2})$.

1.3 Satellites

Two satellites orbit Earth on the same plane along elliptic paths of eccentricities e_1 and e_2 respectively. The angle between their major axes is α . The rate at which a line segment connecting a satellite and the Earth’s centre sweeps out an area is L_1 and L_2 respectively. What is the maximal relative velocity of the satellites? Provide also a simplified answer for $\alpha = 90^\circ$ and $L_1 = L_2$. The ratio of the orbital periods of the satellites is an irrational number. Earth’s mass is denoted by M , and the gravitational constant by G . (Physics Cup 2024, P4)

2 Electromagnetism

2.1 Oscillating Dipole

An electric dipole is placed a distance h away from an infinite metallic plane. The dipole is fixed from its center, but free to rotate. Find the equilibrium positions for the dipole. Determine if these positions are stable, and if so, find the frequency of small oscillations. (Kutay)

2.2 Miscellaneous

2.3 Magnetic Billiards

Consider two absolutely elastic dielectric balls of radius r and mass m one of which carries isotropically distributed charge $-q$, and the other, $+q$. There is so strong homogeneous magnetic field $\vec{B}$, parallel to the axis z , that electrostatic interaction of the two charges can be neglected; neglect also gravity and friction forces. The first ball (negatively charged) moves with speed v and collides with the second ball which had been resting at the origin. The collision is central, and immediately before the impact, the velocity of the first ball was parallel to the x -axis. (NBPhO 2018, P5)

- i) What is the speed of the second ball immediately after the collision?
- ii) Sketch the trajectories of the centres of the both balls during the subsequent motion.
- iii) What is the average velocity (magnitude and direction) of the balls during their subsequent motion?
- iv) Consider the same situation as before, except that there are three differences in the assumptions: the both balls have now identical positive charge $+q$; electrostatic repulsion between the balls is no longer negligible; the collision is not necessarily central (but the balls move still at the same value of z so that the collision will not induce any motion in the direction of the z -axis). Let P_i denote the point where the surfaces of the two balls are in contact during the i -th collision. What is the maximal distance between P_i and P_j (maximize over all the values $i, j = 1, \dots, \infty$, and over all the impact parameters of the collisions for fixed values of B , m , and q)?

2.4 Charge on a Ring

A point-like particle with mass m and charge q is free to slide without friction along a fixed horizontal circular ring with radius r . In the plane of the ring, another charge Q is placed in a fixed position, at a distance d from the center of the ring, with $d < r$ (see figure). (NBPhO 2016, P1)

- i) The particle on the ring is given an initial velocity v_0 . Calculate its velocity v as a function of the angle ϕ , $v(\phi)$.
- ii) How large is the force from the ring on the particle, as a function of the angle ϕ ?
- iii) Viscous friction can be modelled with a force directed against the velocity, with a magnitude proportional to the speed, i.e. $F_f = m\gamma v$, where γ is a positive constant. Assume that this kind of frictional force acts on the particle on the ring. Given an initial v_0 , determine the position where the particle comes to rest.

PROBLEM 1: CIRCLES ALL THE WAY

This problem has two parts.

A. Determining the radius of curvature of a planar curve by means of Mechanics

Each infinitesimal length element of a curve can be thought as belonging to a circle with a certain radius (see **Figure 1**). This radius is called the *radius of curvature*. In order to determine the radius of curvature at some point of a planar curve, one can consider that the curve is the trajectory of a point-like object, and that the given point is the very object.

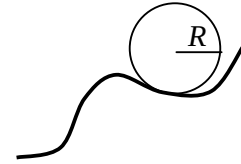


Figure 1

Let $y = f(x)$ be the equation of the curve, where x and y are the coordinates of the object.

- Express the components v_x and v_y of the velocity of the object in terms of x and its time derivatives, and $f(x)$ and its derivatives.
- Let a_t be the component of the acceleration vector parallel to the velocity. Express vector $\vec{a}_t$ in terms of v_x , v_y , the components a_x and a_y of the acceleration, and the unit vectors $\vec{i}$ and $\vec{j}$ of the Ox and Oy axes.
- Express the magnitude (i.e. the absolute value) of the component $\vec{a}_n$ of the acceleration perpendicular to the velocity, in terms of v_x , v_y , a_x and a_y .
- Express the radius of curvature in terms of $f(x)$ and its derivatives.
- Find the radius of curvature of the parabola $y = Ax^2$ ($A > 0$) at a point having coordinate $x = x_0$.
- Find the period of small oscillations performed by a bead on the bottom of a smooth surface of equation $y = \sin 2x$ [m]. You may use $g \approx \pi^2$ [m/s²], where g is the gravitational acceleration.

B. Springs on a circle

In this problem we will investigate the motion of a point-like object of mass m connected to $N + 1$ springs of stiffness k . The springs are attached to a circle of radius R (see **Figure 2**) and have negligible natural length. The points to which the springs are attached are arranged uniformly on the circle such that, when the object is at the center of the circle, the angle between two adjacent springs is equal to $2\pi/(N+1)$. We label the springs 0 through N .

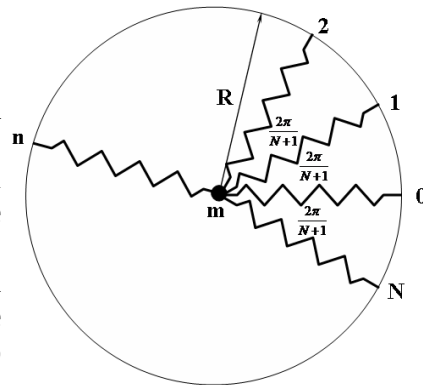


Figure 2

Let angle α and the radial coordinate r of the object be defined as in **Figure 3**. The object can move in the plane of the circle and we ignore the effects of gravity throughout the problem.

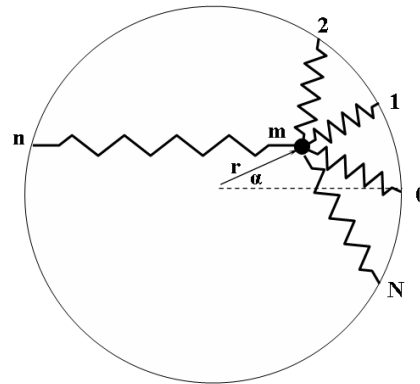


Figure 3

g. Compute the length l_n of the n -th spring for arbitrary r and α .

h. Write down the kinetic energy E_{kin} and the potential energy E_{pot} of the object in terms of r , α , and their time derivatives. (Do not evaluate the sum in the potential energy just yet.)

i. Compute the sums

$$\sum_{n=0}^N \cos\left(\frac{2n\pi}{N+1}\right) \quad \text{and} \quad \sum_{n=0}^N \sin\left(\frac{2n\pi}{N+1}\right)$$

and use the results to evaluate E_{pot} for arbitrary N .

j. Show that the angular momentum L is conserved.

k. Write down the implicit equation for r in terms of $\mathcal{L} \equiv L/m$ and $\omega^2 \equiv (N+1)k/m$.

l. Perform the substitution

$$r(t) = \sqrt{z^2(t) + K},$$

with K an unspecified parameter, and obtain the equation “of motion” for $z(t)$.

Perhaps surprisingly, this equation admits oscillatory solutions $z(t) = A \cos(\omega t + \phi_0)$.

m. Show this, and determine A in terms of $\mathcal{L}$, K , and ω .

K and ϕ_0 can be thought of as integration constants to be specified by the initial conditions. Since the equation of motion is second order, and we have two integration constants, it means that this is the most general function describing the distance $r(t)$. On the other hand, one can notice that the equation for $r(t)$ remains unchanged on substituting r with $-r$. Physically this is the same as substituting α with $\alpha + \pi$, which means that the distance from the center of the circle to the object is the same in any two opposing directions.

n. Write down $r(t)$ in terms of $\mathcal{L}$, ω , K and ϕ_0 . Is $r(t)$ periodic? Is the motion periodic? Do they both have the same period?

o. Describe the motion of the object when $L = 0$.

p. Find out the possible value of r at which the object could perform circular uniform motion.

Suppose now we remove the springs labeled $0, d, 2d, \dots$, where d divides $N+1$.

q. Argue that $r(t)$ derived in part **n.** continues to hold, but for a different value of ω . What is this new value ω' in terms of ω , N , and d ?

2. OSCILLATIONS OF ELASTIC BODIES

In this problem gravitational effects are to be neglected. Unless otherwise stated, substances are to be considered homogenous and isotropic at all times.

A. Consider a very thin elastic rod, prevented from bending. The rod has length L , density ρ , and Young elasticity modulus E . The cross section of the rod is to be taken as constant. The rod is slightly stretched from both ends and then let to oscillate freely.

- Write the kinetic energy of the rod at some moment in terms of its mass m , length L , and change rate of the tensile strain ε .
- Write the potential elastic energy of the rod at some moment in terms of m , ρ , E , and ε .
- Show that the conservation of mechanical energy for the rod leads to the characteristic differential equation for an undamped harmonic oscillation. Resorting to the analogy with a point mass m acted upon by the force σS (σ being the tensile stress and S being the cross section), specify the quantity that plays here the role of coordinate.
- Write the expression for the period of small longitudinal oscillations of the rod in terms of L , ρ , and E .

B. Consider an elastic sphere of radius R , made of the same material as the rod before. Let ε be the tensile strain $\Delta R/R$. The sphere is slightly compressed uniformly and then let to oscillate freely.

- Write the mechanical energy of the sphere at some moment in terms of m , R , ρ , E , ε and the change rate of ε .
- Write down the expression for the period of small radial oscillations of the sphere in terms of R , ρ , and E .

C. Consider a very thin rectangular elastic plate, prevented from bending. The plate has the linear dimensions L and l respectively. The thickness of the plate is to be taken as constant.

The effects of the tensile stresses σ_x and σ_y acting on the plate are **NOT** independent, in the sense that a stretching on one of the directions leads to a shrinking on the other direction. In the limits of Hooke's Law, this can be written as:

$$\begin{aligned}\varepsilon_x &= -\mu \frac{\sigma_y}{E}, \\ \varepsilon_y &= -\mu \frac{\sigma_x}{E},\end{aligned}$$

where the dimensionless factor μ (Poisson's ratio) is somewhere in the range of 0.3.

- Express σ_x and σ_y in terms of ε_x , ε_y , E , and μ .
- Write the system of differential equations for the movement of a point mass m on two orthogonal directions, using as coordinates the equivalent found at point **c**.
- Find the possible values of ω for which the solutions of the above system are simple undamped harmonic oscillations (modes):

$$\begin{aligned}\varepsilon_x &= A \sin \omega t \\ \varepsilon_y &= B \sin \omega t\end{aligned}$$

Express the results in terms of L , l , ρ , E , and μ .

j. In general, the solution of the above system of equations is a superposition of the two modes found. For a square plate ($L = l$) and a weak Poisson ratio ($\mu^2 \ll 1$), express the beats period in terms of μ and the period T_{long} of longitudinal oscillations of a rod with the same length.

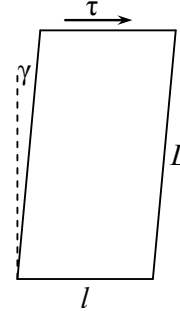
D. Now, instead of being squeezed, the plate is slightly slanted along one of the dimensions, by the action of a shear stress τ , as in the diagram alongside. The shear strain is defined to be $\tan \gamma \approx \gamma$, and Hooke's Law takes the form:

$$\gamma = \frac{1}{G} \tau ; [\tau]_{\text{SI}} = [G]_{\text{SI}} = \text{N/m}^2 ,$$

where G is the so-called *shear modulus*.

k. Express G in terms of E and μ .

l. Find the period of the small slanting oscillations of the plate in terms of L , ρ , and G . Express the same period in terms of μ and the period of a rod with the same length undergoing longitudinal oscillations, T_{long} .



E. A cylinder of radius R and length L , made of the same material as before, is slightly twisted and let to oscillate freely. The torsion strain is defined as the angle θ with which the cylinder is twisted, under the stress of a torque. Hooke's Law takes now the form:

$$\theta = \frac{1}{C} M ; [C]_{\text{SI}} = [M]_{\text{SI}} = \text{Nm} ,$$

where C is the so-called *elastic torsion constant*.

m. Find the period of the small twisting oscillations of the cylinder in terms of L , ρ , and G .

n. Express the elastic torsion constant in terms of R , L , E , and μ .